

ATPL EASA STUDY GUIDE | 010 - AIR LAW

Chapter 11: Parallel Runway Operations

Syllabus Source: CAE Oxford (pp. 239-252) | Pass Target: >= 90% AviationExam

1. MODES OF OPERATION & MINIMUM CENTERLINE SPACING

Simultaneous operations on parallel or near-parallel runways are governed by ICAO Doc 4444 (PANS-ATM) and Doc 8168. Minimum centerline separation distances are strictly tested:

Operational Mode	Min. Centerline Spacing	Operational Criteria & Stagger Requirements
Independent Parallel Approaches	At least 1,035 m (3,400 ft)	Simultaneous approaches where radar separation between aircraft on adjacent ILS localizers is NOT required. Requires dedicated radar monitor controllers.
Dependent Parallel Approaches	At least 915 m (3,000 ft)	Simultaneous approaches where radar separation minima (stagger) between aircraft on adjacent extended centerlines IS required (usually 1.5 or 2.0 NM).
Independent Parallel Departures	At least 1,035 m (3,400 ft)	Simultaneous departures where departure tracks diverge immediately by at least 15° after take-off.
Segregated Parallel Operations	At least 760 m (2,500 ft)	One runway is used exclusively for departures, and the other runway is used exclusively for arrivals.

2. PROTECTED ZONES: NOZ AND NTZ

- Normal Operating Zone (NOZ): An airspace corridor extending from runway threshold to point where aircraft intercepts ILS, within which aircraft are expected to maneuver.
- No Transgression Zone (NTZ): A non-maneuvering corridor of at least 610 m (2,000 ft) width located centrally between the two extended runway centerlines. Penetration by an aircraft requires immediate controller breakout instructions.

[!] EXAM TRAP (AVIATIONEXAM) - NTZ Penetration & Breakout Instructions

When an aircraft is observed penetrating the NTZ, the radar controller immediately instructs the threatened aircraft on the adjacent localizer to execute a BREAKOUT MANEUVER (e.g. 'Turn left immediately heading 270, climb to 3,000 ft'). The pilot must execute the breakout without delay.

3. PRECISION RUNWAY MONITOR (PRM) SYSTEMS

PRM radar systems provide fast radar updates (≤ 2.5 seconds) and high-resolution displays. When PRM is in use, independent parallel approaches can be conducted on runways with centerline spacing as close as 1,035 m without staggered separation.

4. INTERSECTING & CONVERGING RUNWAY OPERATIONS

For operations on intersecting or converging runways, take-off and landing clearances are coordinated so that aircraft flight paths never cross simultaneously. When wake turbulence from an intersecting departure crosses the arrival path, a full 2-minute (or 3-minute for intermediate intersection) separation is applied.

5. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: What is the minimum runway centerline spacing for independent parallel instrument approaches?

- [A]: 760 m (2,500 ft)
- [B]: 915 m (3,000 ft)
- [C]: 1,035 m (3,400 ft)
- [D]: 1,500 m (5,000 ft)

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [C]. Independent parallel approaches require at least 1,035 m (3,400 ft) centerline spacing.

Q2: What is the minimum width of the No Transgression Zone (NTZ) between parallel runway centerlines?

- [A]: 300 m
- [B]: 610 m (2,000 ft)
- [C]: 915 m
- [D]: 1,035 m

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. The NTZ must be at least 610 m (2,000 ft) wide and located midway between the two extended runway centerlines.

Q3: For independent parallel departures, what is the minimum track divergence required immediately after take-off?

- [A]: 10°
- [B]: 15°
- [C]: 30°
- [D]: 45°

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Tracks must diverge by at least 15° immediately after take-off for independent parallel departures.

Q4: What is the minimum runway centerline spacing for segregated parallel operations?

- [A]: 610 m
- [B]: 760 m (2,500 ft)
- [C]: 915 m
- [D]: 1,035 m

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Segregated parallel operations (one runway arrival, one runway departure) require at least 760 m spacing.

Q5: What is the required radar stagger distance between aircraft on adjacent parallel ILS localizers during departures?

- [A]: 1.0 NM
- [B]: 1.5 NM to 2.0 NM
- [C]: 3.0 NM
- [D]: 5.0 NM

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Dependent parallel approaches require a radar stagger of 1.5 NM (or 2.0 NM) between successive arrivals on parallel localizers.

Q6: During independent parallel approaches, what immediate action is required from an aircraft instructed to e

- [A]: Request reason from ATC before turning
- [B]: Disconnect autopilot and comply immediately with heading and climb/descent instructions without delay
- [C]: Continue on localizer until visual contact is made
- [D]: Squawk 7600

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. A breakout instruction is issued when another aircraft penetrates the No Transgression Zone (NTZ). The pilot must comply immediately.

Q7: Under PANS-ATM, what is the maximum convergence or divergence angle for two runways to be classified

- [A]: 5°
- [B]: 10°
- [C]: 15°
- [D]: 30°

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [C]. Non-intersecting runways whose extended centerlines have an angle of convergence or divergence of 15° or less are classified as near-parallel runways.